

Tutorial 3

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Q1. Suppose $L \subseteq \Sigma^*$ is a regular language. We define

$$L^{1/2} = \{x : \exists y, |x| = |y|, xy \in L\}$$

Show that $L^{1/2}$ is a regular language using just the Myhill-Nerode Theorem

Q2. Let L be a regular language with n Myhill–Nerode equivalence classes. Show that the reverse language L^R can have as many as 2^n equivalence classes.

Q3. XYZ decides whether or not she will go to work following the rule that in any 7 consecutive days there are at most 4 working days. A calendar of n days is represented as an n -length word over the alphabet $\{0, 1\}$ where 0 represents a working day and 1 represents a day off.

Prove that the set of all words representing valid calendars forms a regular language over the alphabet $\{0, 1\}$. Also give the minimal automata for this language.

Q4. Consider $\mathcal{L} = \{w \in \{0, 1\}^* \mid w \text{ is not a palindrome}\}$. Use the Myhill-Nerode theorem to show that this language is not regular.

Q5. Draw the minimal DFA for the following languages if they are regular and prove that this is the minimal DFA using Myhill-Nerode theorem. If not, prove so using the Myhill-Nerode theorem.

1. $L = \{a^n \mid n \equiv 0 \pmod{5}\}$
2. $L = \{ww \mid w \in \{0, 1\}^*\}$
3. $L = \{0^i 1^j \mid i \leq j\}$
4. $L = \{w \mid \#_0(w) = \#_1(w) \pmod{3} \text{ and } 101 \text{ is a substring of } w\}$ over $\Sigma = \{0, 1\}$

Q6. Check if the following languages satisfy the Pumping Lemma and the Jaffe's Pumping Lemma to comment on whether the languages are regular:

1. $L = \{a^i b^j c^k d^l \mid i = 0 \text{ or } (i = j \text{ and } k = l)\}$
2. $L = \{w \in \{0, 1\}^* \mid w \text{ contains a cube, i.e., a subword of the form } vvv \text{ where } v \neq \varepsilon, \text{ or } |w| \text{ is not a power of } 10\}$
3. $L = \{w \in \{0, 1, 2\}^* \mid w \text{ contains a square, i.e., a subword of the form } vv \text{ where } v \neq \varepsilon, \text{ or } |w| \text{ is not a power of } 10\}$

Q7. Can you think of additional examples of non-regular languages and show that they satisfy the Pumping Lemma but fail Jaffe's Pumping Lemma?